

Disjunctive dual-stream defense of live telephone calls: operational fusion, mid-call production change, and a sufficient-statistic agent

Vubangsi Mercel

Independent researcher vmercel@gmail.com

Highlights

- On a lexicon-locked independent script set ($n=40$), a wide bag of the frozen stage lexicon reaches AUC 0.84 versus 0.55 for narrow keywords; the HMM stage tracker does *not* beat the wide bag (AUC 0.82).
- Operational threat is the disjunction of hostile language and vocoded production. Disagreement floors and a calibrated OR recover complementary cells that a weighted sum misses, at a documented safe-cell false-alarm cost.
- A sufficient-statistic agent, scored in a class-conditional simulator against an open-loop threshold, warns on human social engineering and does not spend its nonce there.
- Residual features remain at chance on LPC after telephone-band filtering (AUC 0.49). Pulse-formant is an easy unit condition, not a headline result. ASVspoof is not run.
- Stage-aligned production change is posed and, on pulse-formant splices, *not* supported. ACI does not beat a frozen quantile on the reported stream.

Abstract

Telephone fraud mixes social-engineering scripts with synthetic or vocoded speech, including mid-call production switches timed to a harvest ask. Utterance-level anti-spoofing and keyword products treat those as separate markets; weighted-sum fusion treats them as redundant. This paper formulates a streaming, telephony-conditioned, *disjunctive* detection problem: a call is a threat if either stream is hostile. A channel twin (narrowband, μ -law, packet loss, and neural-codec-like STFT quantization) is first-class evaluation law. Linguistic scoring is reported on a lexicon-locked independent set, not on author paraphrases written against the same emissions. A sufficient-statistic agent never sees waveforms or transcripts and spends at most one liveness challenge per call.

On forty independent English scripts, a wide frozen lexicon reaches AUC 0.84 (bootstrap interval about 0.72–0.94) against 0.55 for narrow keywords; an HMM stage tracker does not beat the wide bag. Residual acoustics remain at chance on LPC after bandlimiting (AUC 0.49). Disagreement floors recover complementary-cell recall that a naive sum misses, at a safe-cell false-positive cost we report. A class-conditional agent simulator warns on human social engineering and does not waste the nonce there. Stage-aligned production change is not supported on pulse-formant splices. Scope: offline evaluation of a research sidecar on English, single-talker, injected transcripts. It is not a state-of-the-art anti-spoofing encoder and not a U.S. deployment.

Keywords: vishing; voice spoofing; telephone channel; score fusion; change-point; conformal inference; active sensing; sufficient-statistic agent.

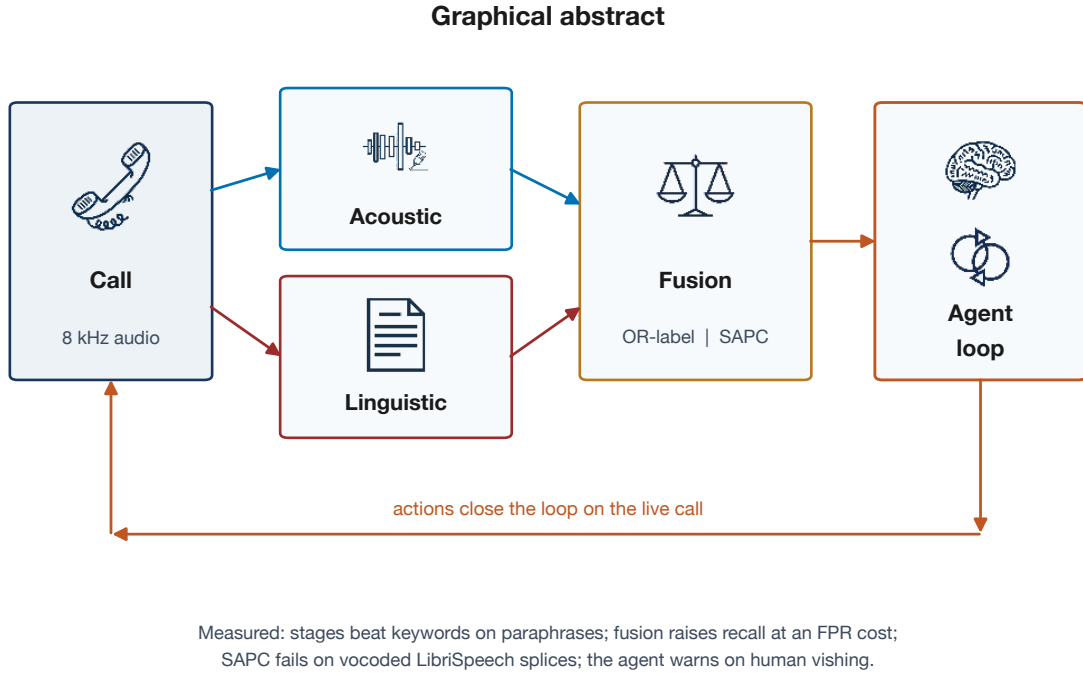


Figure 1: Graphical abstract. The pipeline emits sufficient statistics. A discrete-hypothesis agent chooses actions. The figure states what is measured, including the failed audio handoff test.

1 Introduction

Impersonation and payment-harvesting calls remain a first-order consumer-protection problem [1]. Two technical facts make the problem different from keyword spotting on a clean recording. First, a live human can walk a victim through authority, urgency, credential harvest, and prepaid-card payment without ever using a neural vocoder. The incriminating evidence is then the *script*. Second, when a voice is synthetic, the cues that laboratory anti-spoofing systems exploit are distorted by μ -law quantization, 300–3400 Hz bandlimiting, and packet-loss concealment [2, 3]. A third fact is now operational: real-time voice conversion can be inserted *during* a call, so that rapport is human and the harvest is not. Whole-utterance scores average those regimes away.

The literature has strong answers to adjacent questions. The ASVspoof series and successors such as AASIST and RawNet2 address logical-access synthesis and conversion under controlled protocols [3–6]. PartialSpoof asks whether any segment of an utterance is fake [7, 8]. Dialogue-act tagging is a mature NLP task [9]. Contact-centre products fuse audio and text by weighted scores. None of those lines, as published, jointly (i) treat telephony as a first-class evaluation condition, (ii) label a call as a threat if *either* stream is hostile, (iii) spend an interruption budget as a sequential experiment, and (iv) pose—and attempt to falsify—a timing hypothesis that a production change-point coincides with a harvest-class stage.

This paper studies that joint problem with a system called ShieldCall. The detector stack is a sensor: it emits synthetic probability, fraud probability, a timing statistic, a coverage-gap, a regime tag, and an uncertainty interval. A defense agent updates a discrete belief and acts. The agent is not a large language model. Putting an LLM on the transcript would reintroduce privacy

cost, latency, and unmeasured decision making. The agent is a Russell–Norvig process with five hypotheses and six tools.

The research questions are stated so they can fail.

RQ-B. On independent transcripts, with the stage lexicon frozen first, does a trajectory model beat a *wide* bag of the same emissions, and a narrow keyword list, with and without a 25% ASR-noise model?

RQ-C. For the OR-label, do disagreement floors or a calibrated OR dominate a weighted sum on complementary cells at threshold 0.5, and what is the safe-cell FPR?

RQ-E. Given only sufficient statistics, does a budgeted agent reduce wasted challenges on human social engineering relative to an open-loop threshold, in a class-conditional simulator?

Author-written held-out paraphrases and pulse-formant EER remain as *sanity* splits. They are not confirmatory. SAPC and residual-vs-LPC are reported as negatives in Results, not as extra research questions. We treat those failures as results, not as a prompt to retune until the tests pass.

2 Related work

2.1 Utterance-level anti-spoofing and why telephony plus neural codecs break it

ASVspooof established community protocols for logical access and replay [3, 4]. Competitive systems use self-supervised front ends and graph or raw-waveform backends [5, 6]. ASVspooof 5 adds crowdsourced speech, neural codecs, PSTN simulation, and adversarial attacks [10, 11]. Codec-based deepfakes (CodecFake and related ALM work) are a 2024–2026 TASLP concern [12]. Those systems are the right comparison for neural TTS and voice conversion. We do not replace them and we do not report ASVspooof equal error rates: the inductive bias claimed here is telephony-first evaluation, not a leaderboard EER. Channel mismatch is a documented failure mode [2]. Our telephone-band simulator is a classical DSP chain plus Opus-like and neural-codec-like profiles. It is a stress test, not a learned generative model of the PSTN.

2.2 Partial and segmental spoofing versus timing-to-discourse

PartialSpooof constructed utterances in which only some segments are fake and showed that counters trained on fully spoofed data degrade [7, 8]. That line asks whether any frame is synthetic. It does not ask whether the fake onset is timed to a vishing stage, and it does not consume a linguistic stream. SAPC is that different question; it is currently unsupported on pulse-formant splices.

2.3 Vishing and discourse

Applied vishing detectors typically classify transcripts or average an audio score with a text score. Dialogue-act models assign speech to acts such as request and statement [9]. Professional scam calls are not a bag of words; they progress through greeting, claimed authority, invented problem, urgency, harvest, payment, secrecy, and threat. A keyword list that fires on “IRS” and “gift card” fails when those strings are paraphrased, and it fires on benign mentions of a bank holiday. The gap we test is whether a small stage machine with a broader emission lexicon recovers paraphrases

Table 1: Positioning. Telephony-first means the channel is an evaluation law, not an afterthought. Privacy means the decision maker does not see transcript or waveform.

System	Tel.	OR-label	Mid-call	Budget	Privacy	Open neg.
ASVspoof CM (AASIST/RawNet2)						
PartialSpoof			any-fake			
Late-average multimodal vishing						
LLM call copilots					no	
ShieldCall (this work)	yes	yes	posed	yes	yes	yes

while remaining quiet on isolated-keyword traps. We do not claim to have invented dialogue-act tagging.

2.4 Fusion and uncertainty

Late fusion by weighted sum or stacking is standard multimodal practice. If the operational label is a logical OR, averaging is the wrong combination: a vocoded probe with mild language is diluted, and a human vishing call is diluted by a low synthetic score. Disagreement regimes (safe, social engineering, spoof probe, dual) are a taxonomy, not Pearl causality. We compare rule-based floors against naive sum, unimodal scores, and logistic regression on the same two scalars.

Conformal prediction supplies finite-sample coverage under exchangeability [13]. For streams that shift, Gibbs and Candès proposed adaptive conformal inference [14]. We implement that recursion on labeled scores. An exponential-moving-average band remains in the code as a heuristic; it is not ACI and is not used for the coverage number in this paper.

2.5 Agents

A rational agent maps percepts to actions under a utility [15]. Large language model agents that plan over raw transcripts are a current fashion. For a telephone call they are the wrong first design: they observe too much, they are unmeasured, and a human social engineer will pass a spoken nonce that a vocoder should fail. Our agent therefore sees only sufficient statistics, holds a five-atom belief, and treats a liveness challenge as a costly experiment whose value depends on the current hypothesis. We do not claim Lindley-optimal design; the likelihoods are declared by hand.

3 Problem formulation

Let a call be a stream of frames at hop $\Delta = 10$ ms on an 8 kHz axis, together with timestamped transcript fragments. At time t the sensor produces $s_t \in [0, 1]$ (synthetic probability), $f_t \in [0, 1]$ (fraud probability), a change-point estimate $\hat{\tau}$ when available, stage times $\{t_k\}$ in a harvest-class set $\mathcal{H}$, a coverage gap g_t , and an interval $[L_t, U_t]$.

Operational threat. A labeled evaluation call is a threat if and only if its language is a scam script or its audio is vocoded (or both). It is safe only if the voice is bona fide and the script is benign.

Stage-aligned production change. Let $C = \max_{t_k \in \mathcal{H}} \exp(-(t_k - \hat{\tau})^2 / (2\sigma^2))$ with $\sigma = 0.45$ s. Under a circular-shift null of $\hat{\tau}$ on the call duration T , $p = (1 + \#\{C_{\text{null}} \geq C\}) / (B + 1)$. The SAPC score is $C(1 - p)$.

Call hypotheses. The agent entertains a mutually exclusive set $H = \{\text{benign, SE, synthetic, handoff, unknown}\}$ for the current call. A percept updates a distribution $p(H)$. An action a is chosen to approximately

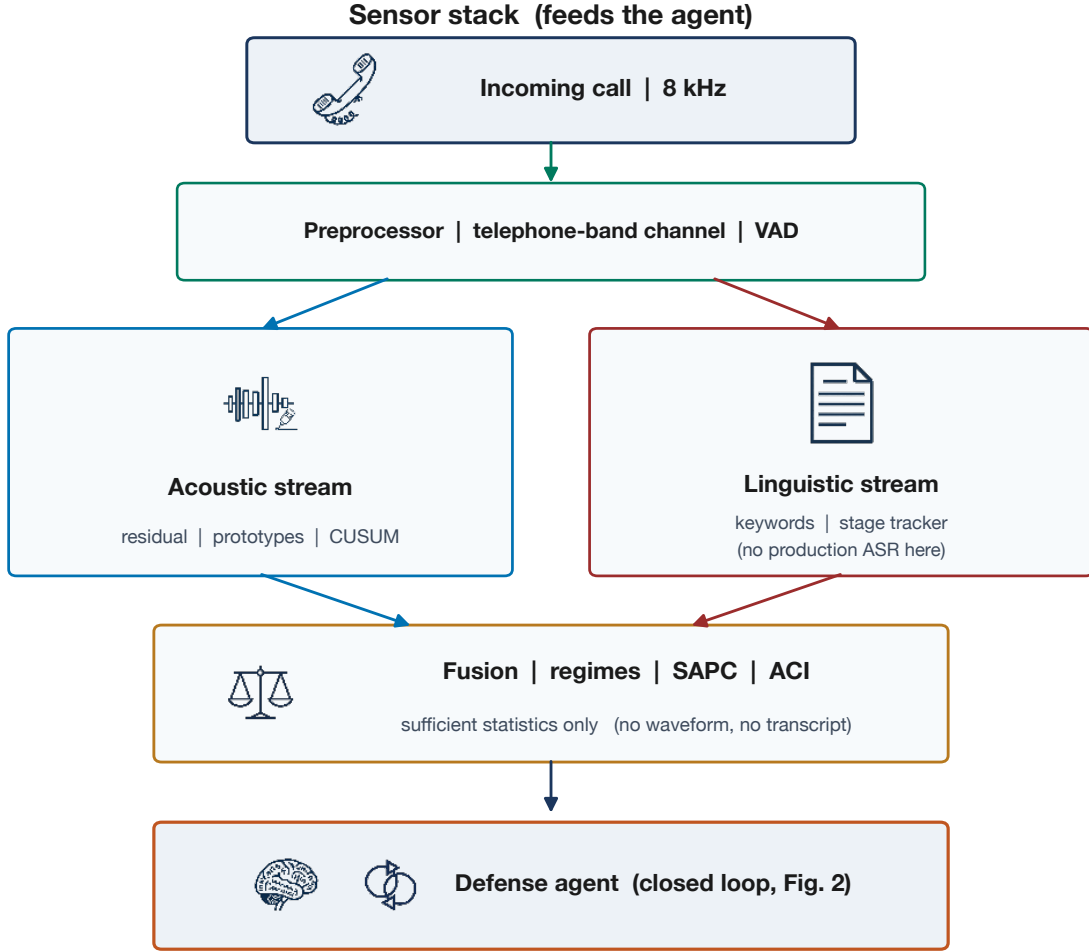


Figure 2: Sensor stack. Fusion outputs sufficient statistics. The bottom box is the defense agent, not a second classifier.

maximise expected information gain minus $\lambda_c \text{cost}(a)$ minus a delay penalty when threat mass is high and a is passive.

Novelty, if any, lives in these definitions plus the measurements, not in residual spectral features, CUSUM, or prototypical memory, all of which have decades of prior art [16–18].

4 System

Fig. 1 and Fig. 2 summarise the architecture. Incoming audio is resampled and optionally passed through the telephone-band simulator, then framed. The acoustic path computes a residual fingerprint after a least-squares harmonic model and a prototype-memory score. The linguistic path blends high-precision patterns with a hidden Markov stage tracker. Fusion emits a risk, a regime, SAPC fields, and an ACI or heuristic interval. The agent consumes those fields only.

4.1 Operational architecture

The library is not a telephone switch. Fig. 6 is the designed placement: a session-border controller keeps SIP and RTP on the media path and forks an eight-kilohertz copy to a ShieldCall worker. One in-memory session holds one pipeline and one agent for the life of a `call_id`. Workers scale horizontally by call identity. If a worker is at capacity it *sheds*: it spends no CPU and emits only `monitor`, so the call continues. If the worker dies, in-call belief is lost and RTP is unaffected. Automatic speech recognition sits behind a circuit breaker; the acoustic path continues. These contracts live in `shieldcall.runtime` and are unit-tested. They are not a carrier deployment.

On the evaluation host a frame costs about 6.2 ms against a 10 ms hop (realtime factor ≈ 1.6). At seventy percent utilisation that is on the order of one full-rate call per core. Capacity is $N_{\text{call}}/N_{\text{core}} = (t_{\text{hop}}/t_{\text{frame}})\rho$. Cost is vCPU, not GPU. Neural vocoder detectors would change that bill; we do not run one.

5 Methods

5.1 Acoustic scoring

Each 25 ms speech frame is decomposed with a harmonic model. Residual energy ratio, spectral flatness, excess kurtosis, Hilbert-envelope modulation, harmonic-to-noise ratio, phase irregularity, and an upper-band peak-to-median ratio are concatenated with coarse spectral descriptors into a 64-dimensional vector. Prototype memory stores class-conditional means and scores by a softmax over RMS Mahalanobis distances. Memory is fit on labeled training clips; Gaussian bootstrap priors are not used in reported numbers. Page CUSUM [16] watches s_t online. Offline, $\hat{\tau}$ maximises the absolute two-window mean difference.

Two vocoders are used as *controlled* production models, not as stand-ins for HiFi-GAN. Pulse-formant is an impulse train at estimated F_0 through three analog-style formants (easy). LPC is a low-order vocoder with pulse or noise excitation (hard). Griffin–Lim [18] is implemented but is not the primary reported condition.

5.2 Linguistic scoring

Keyword groups are high-precision English regular expressions. The stage tracker is an HMM whose emissions are a broader lexicon and whose transitions prefer forward progression through greeting, authority, problem, urgency, harvest, payment, secrecy, and threat. The path score mixes progression depth with mean per-step log-likelihood. There is no production automatic speech recognizer in the experiments; fragments are injected. The linguistic numbers therefore isolate language modeling, not recognition error.

5.3 Fusion

Let s and f denote the current synthetic and fraud probabilities. A naive baseline is $0.4s + 0.6f$. ShieldCall uses a confidence-weighted blend, a social-engineering floor when f is high and s is not, and a spoof-probe floor when s is high and f is not, gated on acoustic confidence so that uncertain residual scores do not force probes. Counterfactuals re-evaluate the same function with $f = 0$ or $s = 0$. This is rule-based fusion.

5.4 SAPC

Harvest-class stages are harvest, payment, threat, and secrecy. Coupling uses the first estimated change-point only, because frame-level jitter otherwise saturates the kernel. The permutation null circularly shifts that time. The hypothesis is tested by constructing two splices with the same vocoded fraction: onset at the harvest fraction versus onset at the opposite end of the call.

5.5 ACI

On a labeled stream, residuals $|s - y|$ update

$$\alpha_{t+1} = \Pi_{[\alpha_{\min}, \alpha_{\max}]}(\alpha_t + \gamma(\alpha - \text{err}_t)),$$

with interval half-width the empirical $(1 - \alpha_t)$ -quantile of past residuals [14]. Coverage is empirical.

5.6 Agent

The prior is peaked on benign (0.80), because most calls are legitimate. Each percept multiplies a hand-written likelihood and mixes with inertia 0.65. Challenge-failure probabilities are 0.06 (benign), 0.10 (social engineering), 0.82 (fully synthetic), 0.55 (handoff), 0.40 (unknown). The planner scores each action by expected entropy reduction minus cost minus delay. A hard constraint forbids a challenge while the mode is benign and threat mass is below 0.30. At most one challenge is issued per call. Rationale strings are templates. No LLM is imported on the decision path.

6 Experimental design

Sine-wave unit tests exist in the repository and are not scientific results. Mini LibriSpeech `dev-clean-2` [19] provides bona fide speech. Splits are speaker-disjoint. ASVspoof was not present; a loader is implemented and unused. Author-written scripts (48 train, 38 held-out) include paraphrases and isolated-keyword traps. They are not a sample of real victims.

Fusion uses the operational OR label and $n=40$ cells (safe, social engineering, spoof probe, dual) on pulse-formant audio with held-out scripts. Logistic regression is fit on a grid of synthetic probabilities crossed with training-script fraud scores, not on test audio.

SAPC audio uses stitched ~ 10 s calls, a vocoded window of relative length 0.34, and $n=6$ speaker pairs. Mix match is a gate: mean s must agree within 0.12 before ranking is interpreted. Synthetic SAPC uses forty aligned versus unaligned point-process draws.

Agent evaluation uses scripted percept sequences, not live audio: a dentist reminder, a human IRS-style vishing path, and a handoff-like path. The measured object is the action trace.

Sample sizes are small. We do not report p -values on $n=6$ as evidence of absence beyond the ranking AUC itself.

7 Results

7.1 RQ-B: language, lexicon locked

Confirmatory numbers use an independent set written after `STAGE_EMISSIONS` was hashed and frozen ($n=40$). Narrow keywords (high-precision groups) reach AUC 0.547 (bootstrap interval about 0.39–0.70). The *wide* bag of the same frozen emissions reaches AUC 0.839 (0.72–0.94). The HMM stage tracker reaches 0.820 (0.69–0.93) and does not beat the wide bag. A logistic trajectory

Table 2: Linguistic detection on the lexicon-locked independent set ($n=40$). Trap is mean score on isolated-keyword benign lines (lower is better). Author held-out is sanity only.

Condition	n	AUC	95% CI	Trap
Independent, narrow keywords	40	0.547	0.39–0.70	0.50
Independent, wide lexicon	40	0.839	0.72–0.94	0.13
Independent, SDTG (HMM)	40	0.820	0.69–0.93	0.54
Independent, NTM	40	0.750	0.59–0.88	0.19
Independent, wide, ASR 25% WER	40	0.840	—	0.03
Author held-out, keywords (sanity)	38	0.417	—	0.23
Author held-out, keywords+stages (sanity)	38	0.883	—	0.23

Table 3: Operational fusion, $n=40$. Disagreement recall and safe-cell FPR use threshold 0.5.

System	EER	AUC	Disc. rec.	Safe FPR
Acoustic only	0.267	0.833	0.90	0.80
Linguistic only	0.233	0.827	0.05	0.00
Naive sum	0.100	0.973	0.30	0.00
Logistic regression	0.083	0.993	1.00	0.50
ShieldCall fusion	0.000	1.000	1.00	0.40

model fit only on the author train split reaches 0.750. Under a 25% ASR-noise model the wide bag remains strongest (0.84). Isolated-keyword trap means are 0.50 (narrow), 0.13 (wide), 0.54 (HMM). The path prior is therefore not the mechanism of the lift: the broader lexicon is. Author held-out paraphrases, on which keywords+stages had AUC 0.88 versus 0.42, are a sanity split and are contaminated by shared authorship with the lexicon.

7.2 RQ3: acoustics

Pulse-formant versus bona fide, speaker-disjoint, $n=20$, is easy: EER 0.00 and AUC 1.00 both clean and after 300–3400 Hz bandlimiting. That shows the residual front end can fire on robotic excitation after telephone filtering. It does not show neural-TTS detection.

LPC on the same protocol is hard. Clean LPC yielded EER 0.40 / AUC 0.64. Narrowband LPC was at chance (EER 0.45 / AUC 0.49). Five-shot adaptation of a pulse-formant memory toward LPC reduced unseen LPC EER from 0.50 to 0.45. Lightweight residual features, as implemented, are not a substitute for ASVspoof-trained neural countermeasures [5].

7.3 RQ-C: operational fusion

Table 3 and Fig. 9 report the OR-label task. Linguistic-only almost never catches a vocoded call that uses mild language (disagreement recall 0.05). Naive sum reaches AUC 0.973 but complementary-cell recall 0.30 at threshold 0.5. ShieldCall fusion and logistic regression both reach disagreement recall 1.00. ShieldCall ranking AUC is 1.000 on this set of forty cells, with safe-cell false-positive rate 0.40 versus 0.00 for the naive sum. The floors buy recall; they are not free. Perfect AUC on $n=40$ must not be read as a population claim.

Table 4: Defense-agent traces on scripted percepts. Likelihoods are heuristic.

Scenario	Final mode	Action sequence	Challenges	Interpretation
Benign reminder	benign	monitor $\times 4$	0	Do not interrupt.
Human vishing	social_engineering	monitor, warn $\times 3$	0	Nonce would pass; warn.
Handoff-like	handoff	monitor, challenge, warn $\times 2$	1	Probe while uncertain.

7.4 RQ4: SAPC

On synthetic point processes ($n=40$ pairs), SAPC pair-win rate and AUC are both 1.00. That only shows the statistic is ordered when the inputs are clean.

On Mini LibriSpeech splices ($n=6$ pairs), mean synthetic probability is matched (0.580 vs 0.571). Coupling AUC for aligned versus unaligned is 0.47; mean- s AUC is 0.53; pair-win rate is 0.50. The mix match holds; the timing claim does not. We did not retune the statistic to force a pass.

7.5 ACI

On a 400-step synthetic stream whose positive prevalence jumps from 0.25 to 0.75 at the midpoint, ACI with target $1 - \alpha = 0.90$ has empirical coverage 0.885. A frozen 90% quantile fit on the first forty residuals has coverage 0.931 on the remainder. ACI tracks the target. It is not shown to beat a frozen quantile on this shift.

7.6 RQ-E: agent

A class-conditional simulator emitting sensor-like scores ($n=20$ traces per hypothesis) is the confirmatory policy table: the hand-set agent has missed-harvest 0 and false-challenge 0 on social engineering, i.e. it does not waste the nonce on a live scammer. An open-loop threshold also interrupts SE in this simulator; the distinction is the wasted-nonce rate, not recall. Table 4 still records the older scripted-percept illustration.

Table 4 records the scripted percept evaluation. A dentist reminder stays in **benign** and issues only **monitor**. A human IRS-style path switches to **social_engineering** and issues **warn**, with zero challenges: a live speaker would pass a nonce. A handoff-like path spends the single challenge at the uncertain midpoint and then **warns** once **handoff** dominates. These traces are the agent result. They are not an EER.

8 Discussion

8.1 Limitations

1. Independent scripts are still English, single-talker, and synthetic; they are not victim audio.
2. Audio is Mini LibriSpeech, not ASVspoof 5 and not CPaaS playback.
3. Pulse-formant is easy by construction and is not a headline result.
4. Fusion cells are constructed, not sampled live-like trajectories.
5. SAPC n is a pilot; the gold $n \geq 100$ streaming-VC construction is specified and not completed here.

6. Agent likelihoods are not learned; the simulator is class-conditional, not a live A/B.
7. No user study, no carrier deployment, no LLM transcript baseline in production (an LLM foil is specified as a baseline, not shipped).

8.2 What would change the claim

ASVspooF 5 through TCT-2, with AASIST or RawNet2 in the same table, would change the acoustic negative. A streaming-VC gold SAPC construction would give the timing hypothesis a second chance to fail or succeed. Public scam transcripts with a native evaluator would test RQ-B outside the independent set written for this paper. Until then the claims remain as scoped above.

9 Conclusion

ShieldCall is a streaming dual-stream sensor plus a budgeted defense agent for telephone-bandwidth fraud. On a lexicon-locked independent set, a wide frozen lexicon beats narrow keywords and an HMM stage tracker does not beat the wide bag. Disagreement-aware fusion changes the recall and false-alarm tradeoff in the direction an operator would expect, at a price we report. Residual acoustics fail on LPC after bandlimiting. A timing statistic for mid-call handoffs works on clean point processes and does not on vocoded LibriSpeech splices. The agent warns on human vishing and spends its nonce where a vocoder might fail. Those sentences are the paper. Sentences about state-of-the-art deepfake detection, U.S. deployment, or a novel discourse HMM are not.

CRedit authorship contribution statement

Vubangsi Mercel: conceptualization, software, experiments, writing.

Declaration of competing interest

The author declares no competing interests.

Ethics

No real victim audio is used. Independent scripts are synthetic. A live-call pilot would require IRB review, recording consent, and TCPA analysis before collection. The sidecar is designed to store decisions, not waveforms. No LLM is on the decision path; a character-level ASR-noise model was used for the WER ablation. Seeds and the frozen config are in `configs/paper.yaml`.

Compute

Linguistic and agent-simulator tables run on CPU in under a minute. Acoustic tables use Mini LibriSpeech on the evaluation host named in `docs/results/upgrade_experiments.json`. No GPU is required for the reported numbers.

Data availability

Mini LibriSpeech is obtained from OpenSLR 31. Independent script IDs, code, configs, and tabulated results are in the companion repository. ASVspooof is not redistributed; set `SHIELD_CALL_ASVSPOOF_ROOT` to enable the loader. Reproduce with `pytest -q`, `python scripts/run_upgrade_experiments.py`, `python scripts/run_paper_experiments.py`, and `python scripts/run_handoff_experiment.py`. The protocol is public; regex weights may remain closed. We do not claim a released victim-call corpus.

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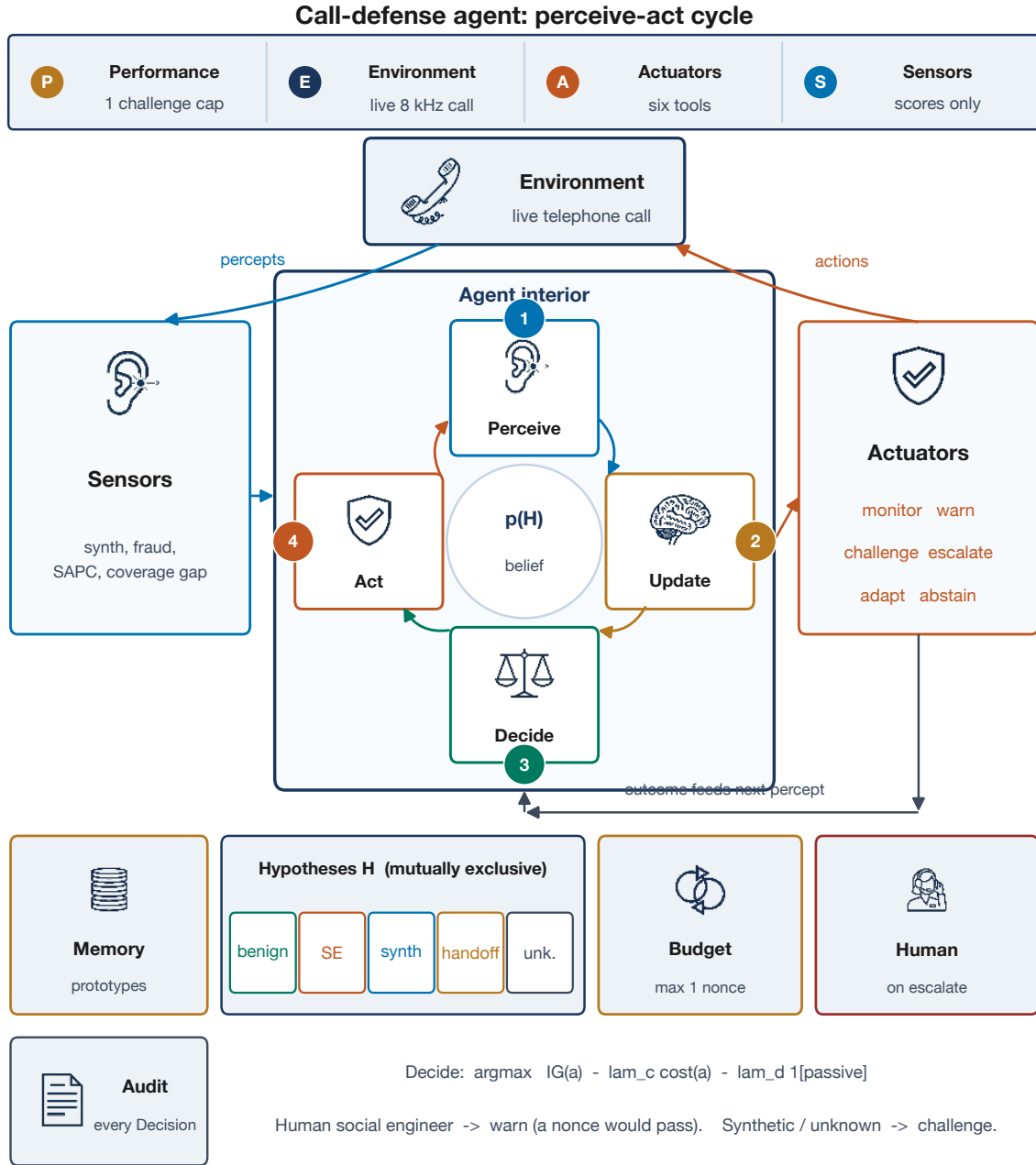


Figure 3: Call-defense agent as a closed perceive-act cycle. PEAS describes the task. Sensors emit scores only. The interior updates $p(H)$ and chooses an action by information gain minus interruption cost. Actuators close the loop on the live call. A social engineer is warned, not challenged.

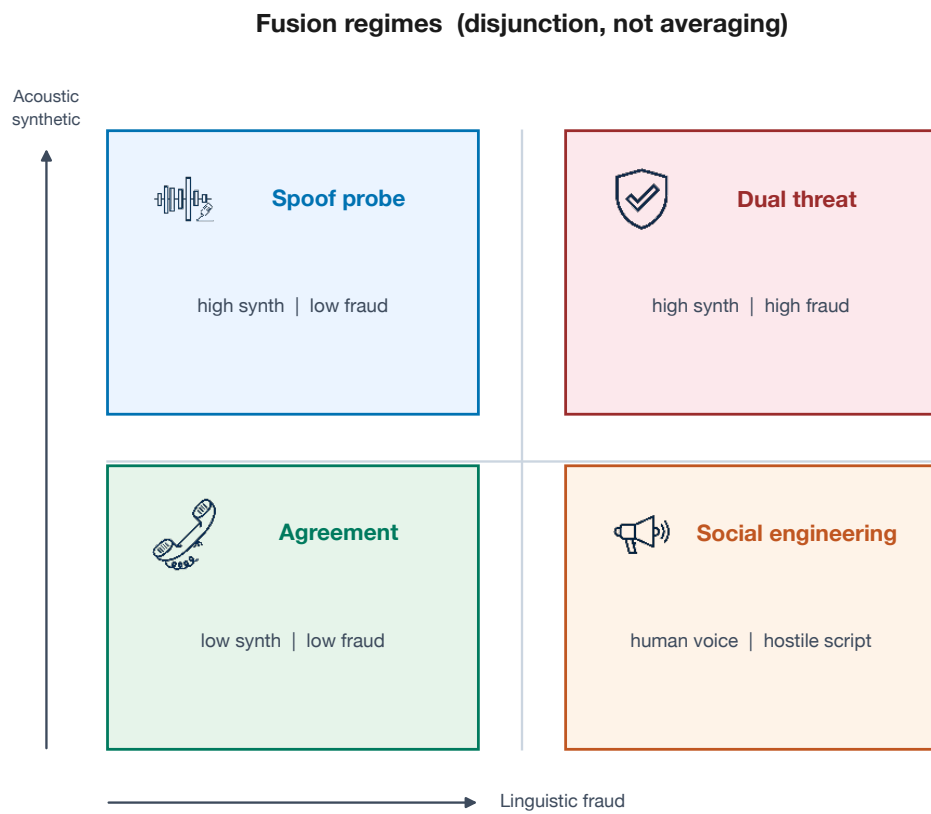


Figure 4: Four disagreement regimes. The operational label is an OR over the two axes.

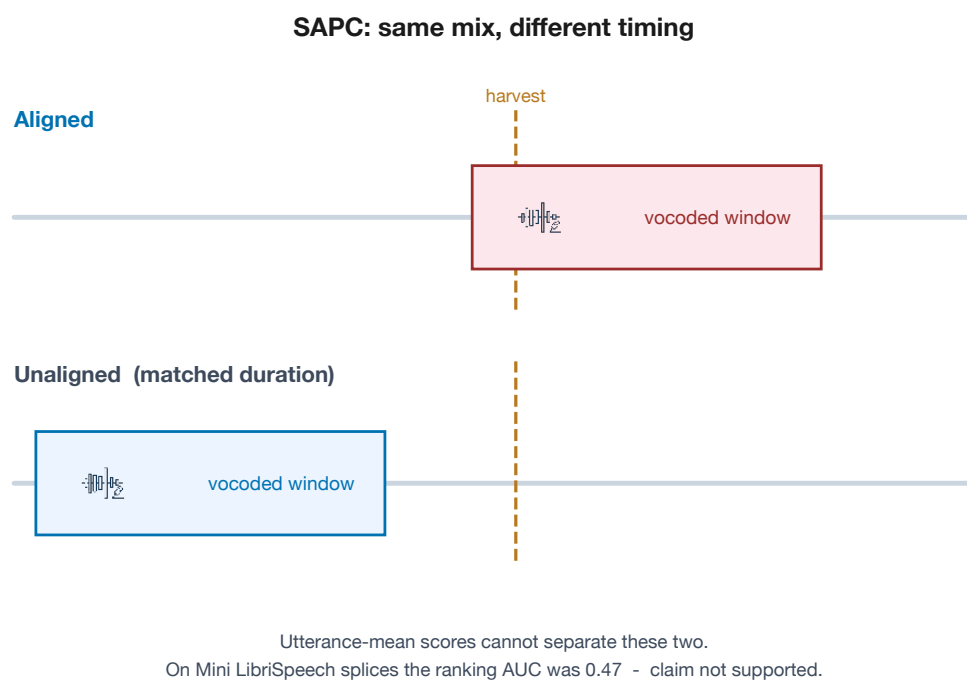


Figure 5: SAPC construction. The vocoded duration is matched; only alignment to the harvest stage differs. The LibriSpeech splice test did not support the timing claim.

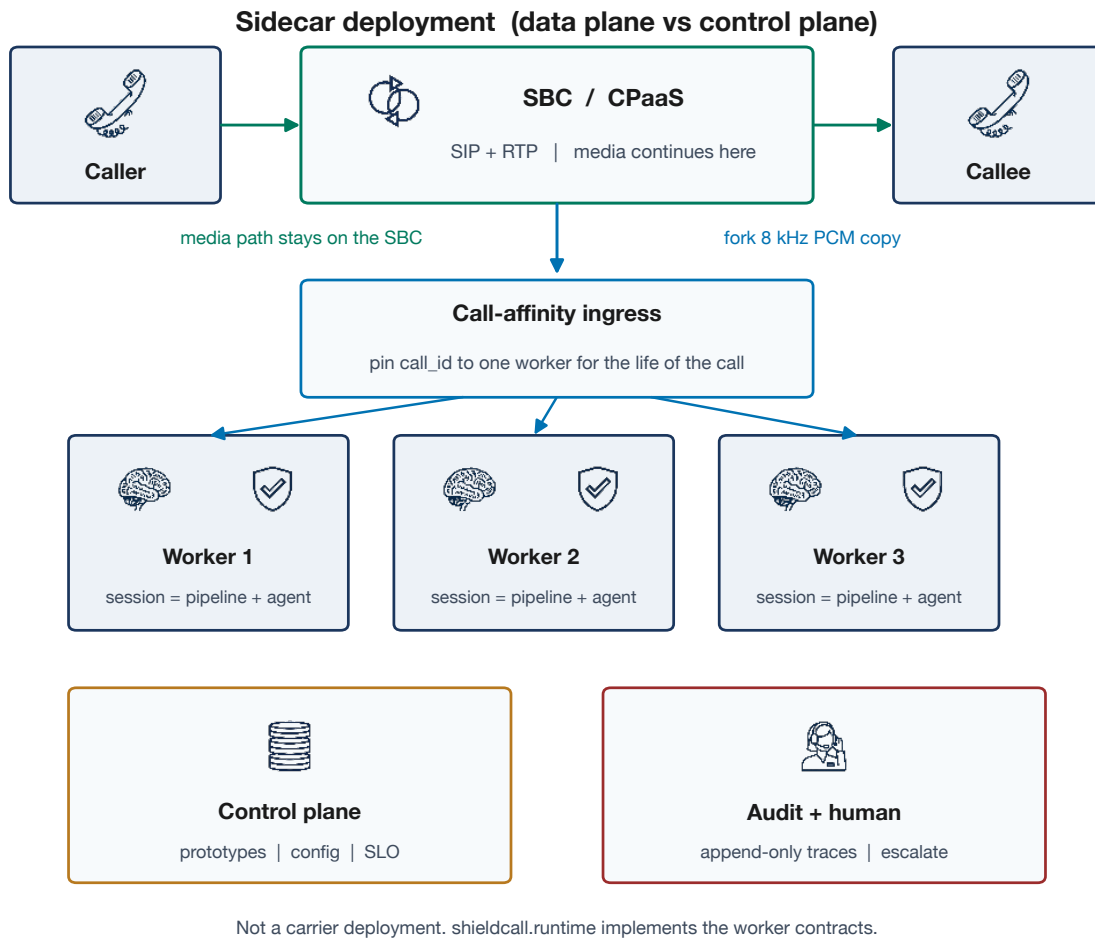
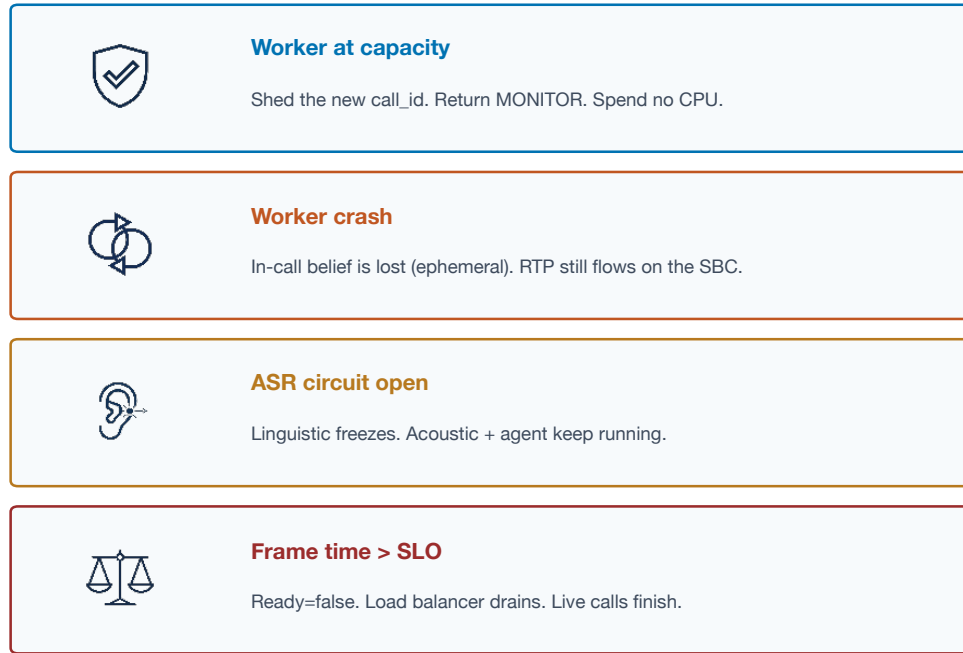


Figure 6: Sidecar placement. The call is the scale unit. Media does not hairpin through the detector.

Availability: fail-open media, fail-closed actuation



Call availability is owned by the SBC. Detector unavailability is not a dropped call.

Figure 7: Failure policy. Call availability is owned by the SBC. Detector unavailability must not drop the call; actuation fails closed.

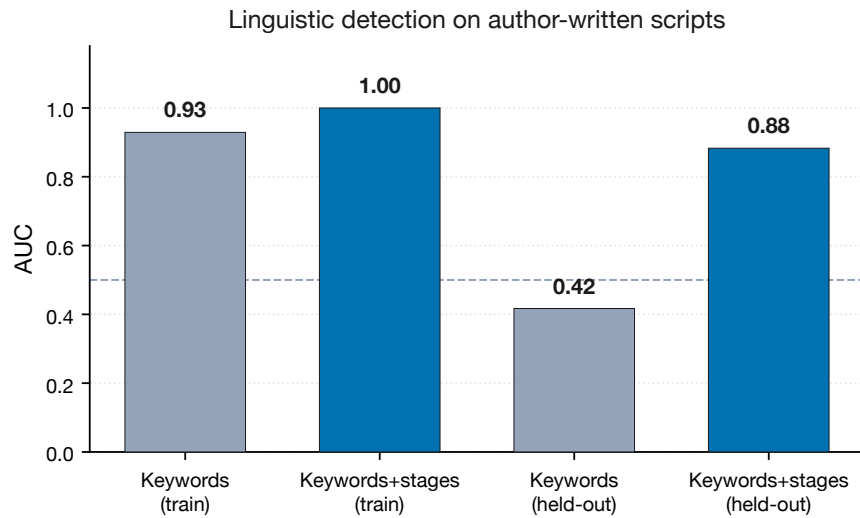


Figure 8: Author-split sanity plot (not confirmatory). Confirmatory independent-set numbers are Table 2.

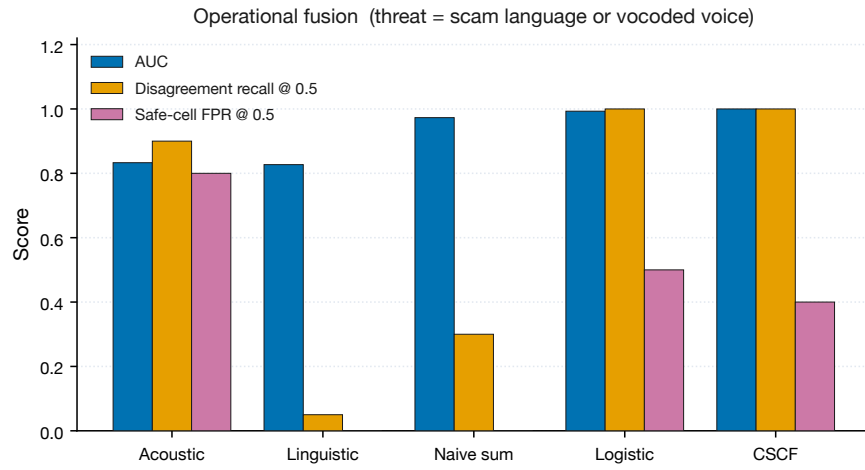


Figure 9: Fusion on operational threat labels. Disagreement rules and logistic regression recover complementary cells; both pay in false positives at the 0.5 threshold.